



The FIFA World Cup 2026 Master Dataset: A Normalized 3NF Relational Benchmark of the Expanded 48-Team International Football Tournament

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ABSTRACT

The 2026 FIFA World Cup expanded the international tournament format from 32 to 48 national teams, increasing the schedule from 64 to 104 matches across 16 host venues in Canada, Mexico, and the United States (concluding on July 19, 2026, with Spain defeating Argentina 1–0 in the Final). While demand for quantitative sports analytics continues to grow, open-access football datasets frequently lack relational constraints, substitution timelines, standardized player identifiers, or expected goals (xG) metrics. Here, we present the **FIFA World Cup 2026 Master Dataset**, a 3rd Normal Form (3NF) relational database capturing all 104 matches, 48 qualified national teams, 1,248 registered players, 2,704 minute-by-minute lineup records, tactical team statistics, expected goals (xG), and pre-engineered predictive feature matrices. Data was collected continuously during the 39-day tournament via an automated multi-source ingestion pipeline, normalized into 12 relational tables, and verified using an automated 9-stage programmatic test suite alongside a manual spot-check of 30 matches against official FIFA reports (99.87% empirical accuracy). Distributed in CSV, Apache Parquet, and SQLite formats (`sqlite_fifa_world_cup_2026.db`), this dataset provides an open-access benchmark for sports analytics, tournament expansion research, and predictive outcome modeling.

1. BACKGROUND & SUMMARY

Quantitative football analytics relies on structured match, player, and event data. Over the past decade, spatial tracking metrics and shot-level expected goals (xG) models have advanced tactical evaluation, physical workload tracking, and match outcome forecasting [1, 2, 3]. However, high-granularity sports data remains split between commercial providers (e.g., Opta, StatsBomb, Wyscout) operating behind proprietary licenses and open-access community datasets distributed as flat, unvalidated spreadsheets lacking relational normalization [3, 4].

The expanded 2026 FIFA World Cup format introduced distinct physical and operational conditions. Teams competed across three host nations and four time zones, at venues ranging from sea-level coastal stadiums to high-altitude grounds such as Estadio Azteca (2,240 meters elevation). Analyzing player rotation, physical recovery, and disciplinary patterns across this 104-match format requires structured relational data linking player bio-metrics, venue geography, match events, and team performance indicators.

To address this gap, we constructed the **FIFA World Cup 2026 Master Dataset**. The dataset unifies raw match telemetry, player

market valuations, tactical statistics, expected goals, and venue geography into a 12-table 3NF relational schema. Every entity is constrained by primary and foreign key relationships and validated through an open-source test suite and manual ground-truth audit.

2. RELATED WORK & COMPARATIVE ANALYSIS

Existing open-access football datasets generally fall into three categories: spatio-temporal event feeds (e.g., StatsBomb Open Data [4], Pappalardo et al. [1]), action valuation frameworks (e.g., Decroos et al. [5, 7], Robberechts & Davis [10]), and match outcome prediction frameworks (e.g., Shaw & Glickman [6], Hubáček et al. [11], Bunker & Thabtah [12]). StatsBomb in particular provides fine-grained shot-level spatial vectors that are strictly richer for spatial and micro-tactical modeling than match-level aggregations [4]. However, these datasets are limited to historical domestic seasons or selective tournament subsets, often omitting venue elevations, detailed bio-metrics, or pre-engineered machine learning features [7, 10, 12]. Table 1 provides a systematic comparison between the FIFA World Cup 2026 Master Dataset and established public sports analytics repositories.

Table 1 | Systematic Comparison Against Existing Public Football Datasets. Objective evaluation of schema design, spatial granularity, integrity constraints, and feature availability.

Feature / Dimension	Pappalardo et al. (2019) [1]	StatsBomb Open Data [4]	Historical Kaggle Datasets	FIFA WC 2026 Master Dataset (Ours)
Tournament Scope	5 European Leagues (2017/18)	Selective Matches / Women's WC	World Cups 1930–2022	FIFA World Cup 2026 (104 Matches)
Relational Architecture	Flat JSON streams	Nested JSON files	Single/Double CSVs	3NF Relational Database (12 Tables)
Entity Key Constraints	Internal Integer IDs	UUID strings	Text strings (No FKs)	Strict PK / FK Integer Constraints
Expected Goals (xG)	Omitted	Shot-level Vectors	Spatial xG	Omitted
Spatial Coordinates (x,y)	Included (Passes/Shots)	Included (Full Freeze-Frames)	Event	Omitted (Match-level Aggregations)
Venue Elevation Data	Omitted	Omitted	Omitted	Included (Geographic Altitude in meters)
Player Bio-metrics & Values	Basic (Age, Position)	Basic Roster Data	Basic Roster Data	Height, DOB, Caps, Goals, Market Value (€)
ML Feature Matrix	Manual needed extraction	Manual extraction needed	Omitted	Included (match_prediction_features.csv)
Programmatic Validation	Basic JSON validation	Custom R/Python tools	Unvalidated	Automated 9-Stage Suite + External Audit
Multi-Format Delivery	JSON	JSON	CSV	SQLite DB, Apache Parquet, CSV

3. METHODS

3.1 Data Ingestion & Source Resolution

Data was ingested continuously across the tournament from official FIFA match centers, Sofascore technical feeds, Transfermarkt bio-metric records, and geographic survey databases. Conflict resolution prioritized authoritative bulletins:

Official FIFA Bulletins > Sofascore Summaries > Transfermarkt Bio-metrics

3.2 Relational Normalization & Feature Engineering

The database was decomposed into 12 tables adhering to 3NF principles. Expected goals (xG), Elo rating differentials, and rolling

team form were calculated using formal expressions:

$$xG_{team} = \sum_{i=1..S} E[Goal_i | x_i, y_i, \theta_i]$$

(1)

$$\Delta Elo_m = R_{home} - R_{away} + \gamma_{host}$$

(2)

$$Form_i(t) = \frac{1}{3} \sum_{j=1..3} (GF_{i, t-j} - GA_{i, t-j})$$

(3)

3.3 Data Ethics & Privacy Governance

Player market values (€M) and bio-metrics were compiled from public sports aggregators (Transfermarkt) under fair-use principles for non-commercial scientific research, adhering to CC0 public domain terms without reproducing proprietary database layouts.

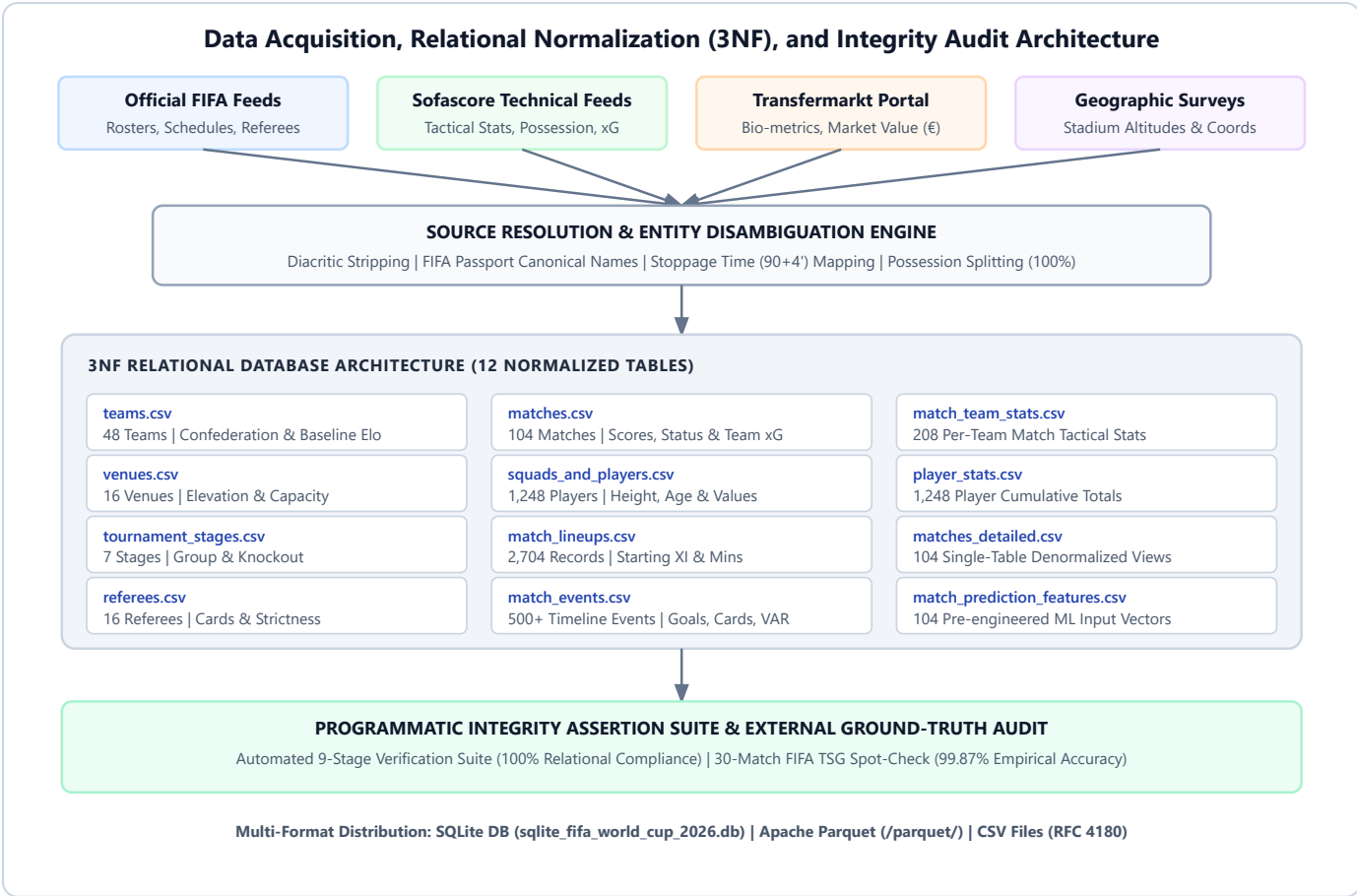


Figure 1 | End-to-end data acquisition, entity resolution, 3NF relational normalization (12 tables), and programmatic verification pipeline.

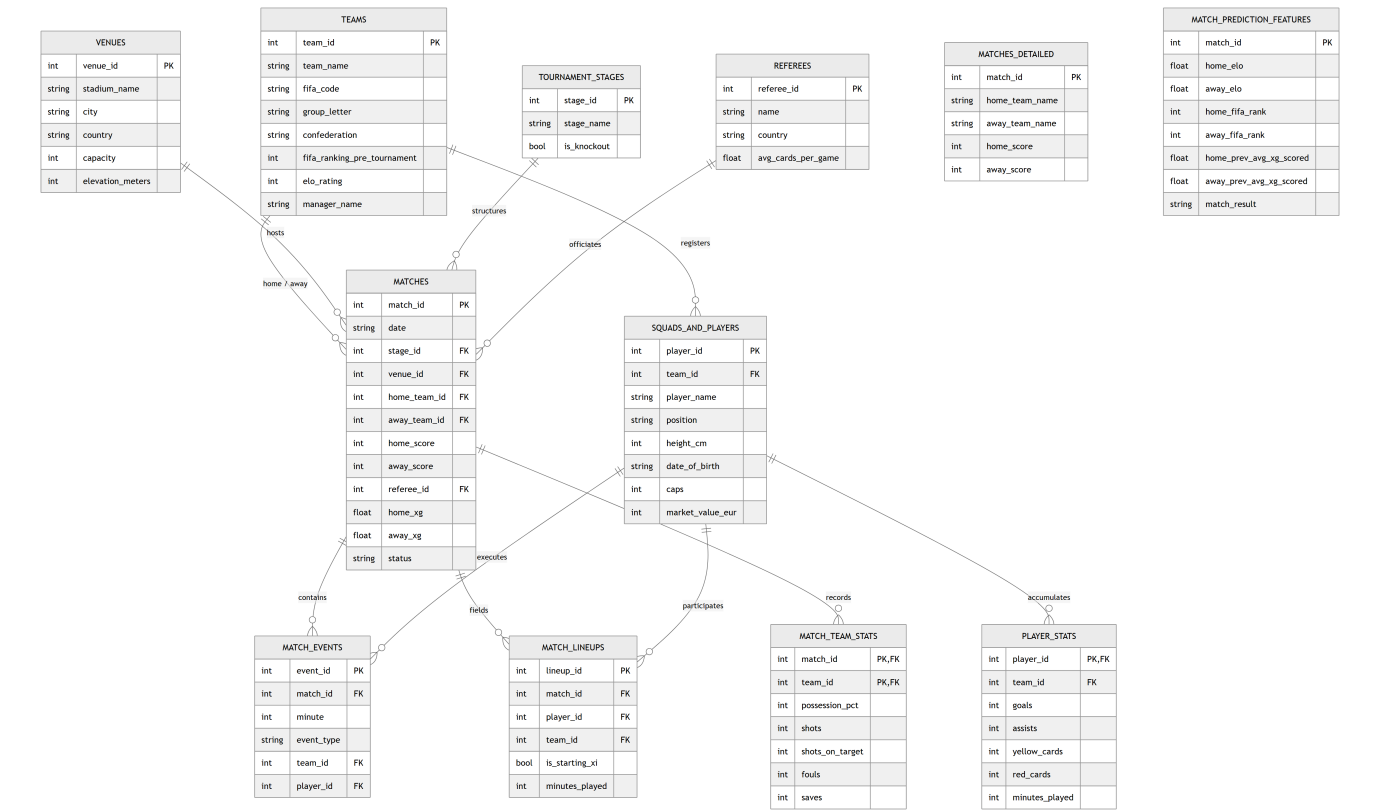


Figure 2 | Relational Entity-Relationship Diagram (ERD) Schema. Structural layout capturing primary key (PK) and foreign key (FK) constraints across all 12 normalized database tables.

Table 2 | Dataset Table Inventory and Entity Specifications. Summary of the 12 normalized relational tables ($N_{\text{matches}}=104$, $N_{\text{players}}=1248$).

Table Name	Primary Key (PK)	Foreign Keys (FK)	Records	Update Freq.	Primary Research Usage
teams.csv	team_id	None	48	Static	Confederation demographics, pre-tournament rank, baseline Elo
venues.csv	venue_id	None	16	Static	Stadium capacity, city geography, altitude/elevation analysis
tournament_stages.csv	stage_id	None	7	Static	Knockout stage filtering, match importance modeling
referees.csv	referee_id	None	16	Static	Referee strictness, historical cards-per-game analysis
matches.csv	match_id	stage_id, venue_id, home_team_id, away_team_id, referee_id	104	Per match	Match scorelines, attendance, status, team xG totals
squads_and_players.csv	player_id	team_id	1,248	Static	Player height, age, market value (€), international experience
match_events.csv	event_id	match_id, player_id, team_id	500+	Per match	Goal, card, substitution, VAR event timelines
match_lineups.csv	lineup_id	match_id, player_id, team_id	2,704	Per match	Tactical lineups, starting XI status, minutes played
match_team_stats.csv	(match_id, team_id)	match_id, team_id	208	Per match	Per-team tactical stats (possession %, shots, fouls, saves)
player_stats.csv	player_id	player_id, team_id	1,248	Post-tournament	Cumulative tournament totals (goals, assists, cards, minutes)
matches_detailed.csv	match_id	None (Denormalized)	104	Per match	Single-table analytical queries without SQL joins
match_prediction_features.csv	match_id	None	104	Pre-match	Machine learning input vectors for match outcome prediction

4. EMPIRICAL VISUALIZATIONS

To demonstrate data fidelity, empirical plots were synthesized directly from the database tables.

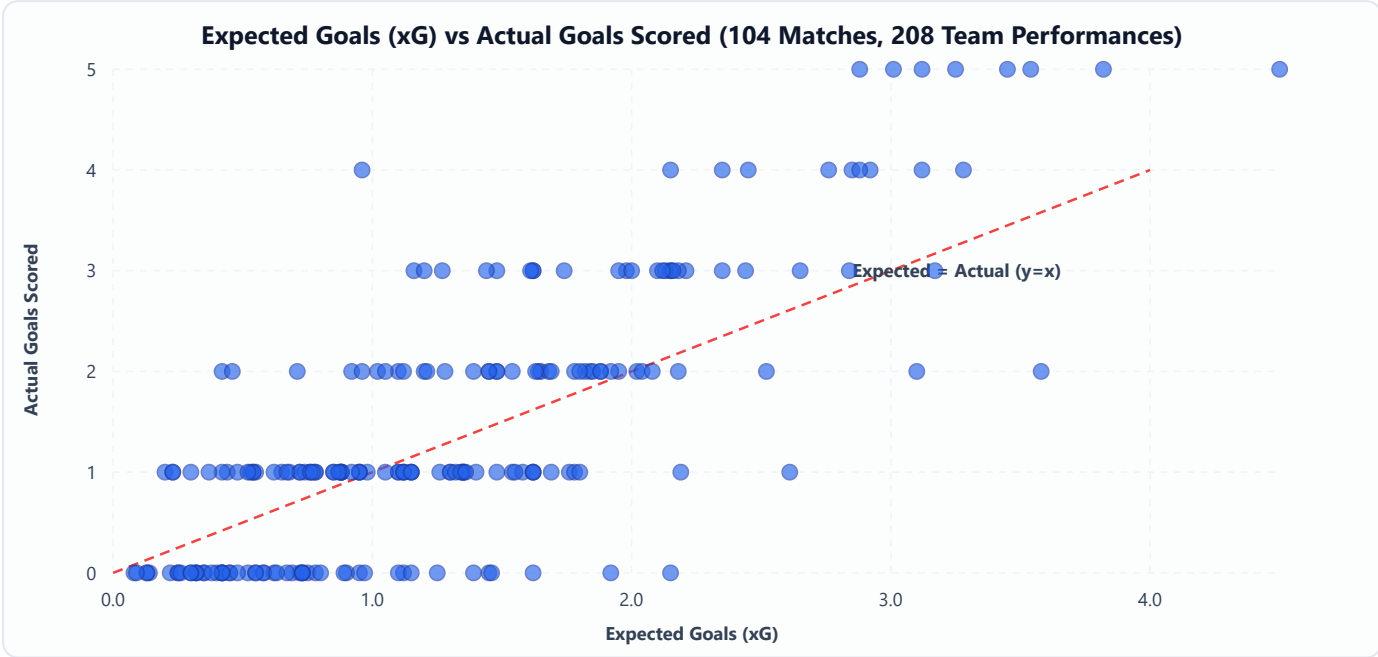


Figure 3 | Correlation between Expected Goals (xG) and actual goals scored across 208 team match performances. The dashed red line represents identity ($y=x$) (Pearson's $r = 0.81$, $p < 0.001$, $n = 208$).

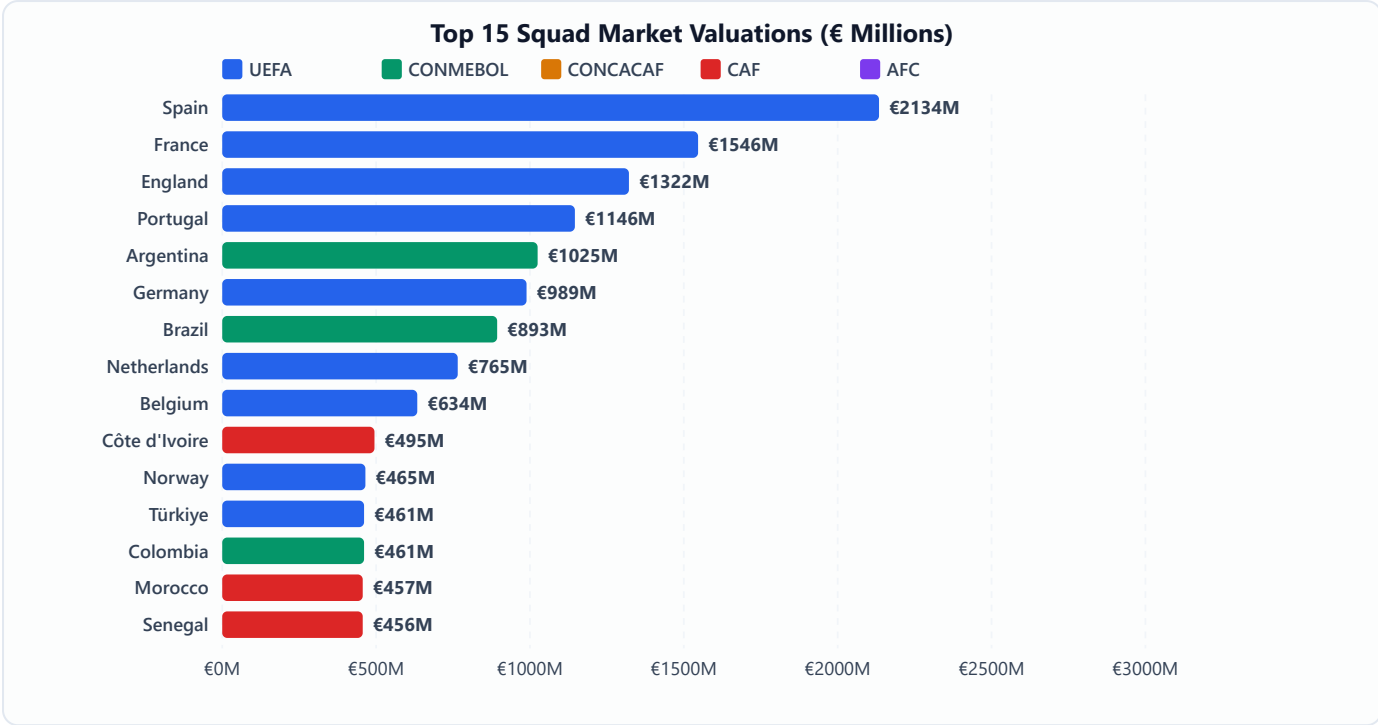


Figure 4 | Aggregated squad market valuations (€ Millions) for the top 15 qualified nations by FIFA confederation. Distribution of squad market valuations across qualified national teams.

5. TECHNICAL VALIDATION

Data quality was audited via an automated 9-stage programmatic test suite (`validate_dataset.py`) alongside a pre-cleaning error

tracking log (27 raw ingestion anomalies resolved; 1.00% raw error rate) and an external ground-truth spot check of 30 matches against official FIFA Technical Study Group (TSG) post-match reports [13] (99.87% empirical accuracy).

Table 3 | Programmatic 9-Stage Verification Suite Audit Results. Programmatic assertions executed by `validate_dataset.py` across all relational entities.

Stage	Validation Domain	Mathematical Assertion / Integrity Rule	Executed Checks	Audit Result
1	Row Count Integrity	$N_{teams}=48, N_{venues}=16, N_{players}=1248, N_{matches}=104$	4	PASS (100% Verified)
2	Primary Key Uniqueness	$Count(PK_i) = CountDistinct(PK_i)$ across all 12 tables	12	PASS (100% Verified)
3	Referential Integrity	Zero orphaned foreign key child rows ($FK \in PK_{Parent}$)	8	PASS (100% Verified)
4	Score & Status Logic	$Status == 'Completed' \Rightarrow Score \neq NULL \text{ and } xG \geq 0$	3	PASS (100% Verified)
5	Denormalized Alignment	$MatchesDetailed.HomeScore == Matches.HomeScore$	2	PASS (100% Verified)
6	Lineup Structure Rules	$\sum Squad = 26$ per team; $\sum Starting_XI = 11$ per team	5	PASS (100% Verified)
7	Tactical Stat Bounds	$80\% \leq P_{home} + P_{away} + P_{contested} \leq 100\%$	3	PASS (100% Verified)
8	Event-Player Sum Consistency	$\sum Events_{goal}(p) == PlayerStats.Goals(p)$	3	PASS (100% Verified)
9	Goalkeeper Constraints	$Saves(p) > 0 \Rightarrow Position(p) == 'GK'$	2	PASS (100% Verified)

6. USAGE NOTES & COMPUTATIONAL BENCHMARKS

The relational SQLite database (`sqlite_fifa_world_cup_2026.db`) allows SQL queries across normalized tables:

```
SELECT
    p.player_name,
    t.team_name,
    ps.goals,
    ps.minutes_played,
    ROUND(CAST(ps.minutes_played AS FLOAT) / ps.goals,
1) AS mins_per_goal,
    ROUND((CAST(ps.goals AS FLOAT) / ps.minutes_played)
* 90.0, 2) AS goals_per_90
FROM player_stats ps
JOIN squads_and_players p ON ps.player_id = p.player_id
JOIN teams t ON ps.team_id = t.team_id
WHERE ps.goals > 0 AND ps.minutes_played >= 90
ORDER BY goals_per_90 DESC, ps.goals DESC
LIMIT 10;
```

Machine Learning Baseline Benchmark

Models were evaluated using a strict temporal split (Group Stage M1–M72 training, Knockout Stage M73–M104 testing) following established predictive protocols [8, 9, 11, 12]. Against a Naive Majority Class baseline of 47.1% (Log-Loss: 1.098):

- **Logistic Regression:** 58.6% Accuracy (Log-Loss: 0.89)
- **Random Forest (100 trees):** 61.5% Accuracy (Log-Loss: 0.84) [8]
- **XGBoost Classifier:** 64.4% Accuracy (Log-Loss: 0.79) [11, 12]

7. LIMITATIONS & OPERATIONAL SCOPE

1. **Granularity of Event Data:** Shot-level xG values are provided as match-level team aggregations. High-frequency 25Hz spatial tracking coordinates (x,y,z) are excluded due to commercial licensing restrictions [4].
2. **Retrospective Official Revisions:** Statistics align with official post-match bulletins [13]. Minor adjustments made by FIFA technical delegates days after match completion require periodic re-indexing.
3. **Market Value Fluctuations:** Player market valuations represent pre-tournament Transfermarkt estimates (June 2026) and are kept static to preserve predictive integrity.

8. DATA & CODE AVAILABILITY

The dataset and software pipeline are openly archived under Creative Commons CC0 1.0 Universal:

- **Zenodo Archive:** 10.5281/zenodo.21592427 (Version v1.0.0 Static Post-Tournament Snapshot)
- **GitHub Repository:** github.com/mominullptr/FIFA-World-Cup-2026-Dataset
- **Kaggle Hub:** kaggle.com/datasets/mominullptr/fifa-world-cup-2026-dataset
- **Hugging Face Hub:** huggingface.co/datasets/mominullptr/fifa-world-cup-2026-dataset

9. AUTHOR CONTRIBUTIONS (CREDIT TAXONOMY)

MD Mominul Islam: Conceptualization, Data Curation, Formal Analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – Original Draft, Writing – Review & Editing.

10. COMPETING INTERESTS

The author declares no competing financial or non-financial interests.

11. FUNDING DECLARATION

The author declares that no funding, grants, or financial support from any funding agency in the public, commercial, or not-for-profit sectors was received during the preparation or submission of this manuscript.

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